Project Description

Project

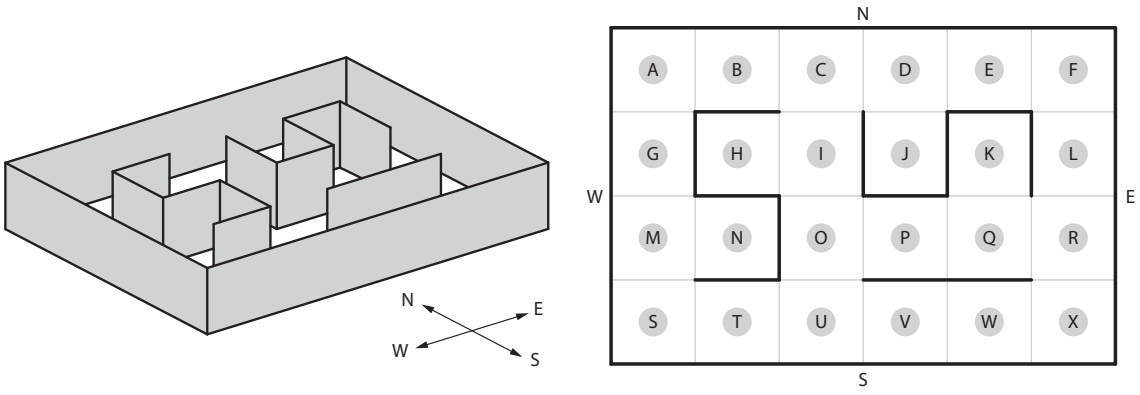
The project will be done in groups of 2-3 students per group. It involves designing robot navigation algorithms through different way points in an environment, while avoiding obstacles, known or unknown. The environment may be changed relative to mapping (e.g., with new static obstacles or missing obstacle/landmarks) and may have dynamic obstacles. Each group will be given access to the following resources:

- Mobile Robot - Waveshare Wave Rover Flexible and Expandable 4WD Mobile Robot Chassis, Full Metal Body, Multiple Hosts Support, with Onboard ESP32 Module
- Edge computing engine: RUBIK Pi 3, the Affordable Edge AI Dev Kit, Built on the Qualcomm Dragonwing™ QCS6490
- Acer Aspire Lite Laptop with ROS

Group assignment and Robot and laptop distribution will occur by the end of Week 4. Please do not modify the internal code of the robot without permission. Unauthorised changes may cause system issues and require staff to spend many hours troubleshooting and recalibrating. This will cause significant delays and hinder your project progress. Please take care to protect the hardware. LiDAR, cameras, and other sensors are delicate. If damaged, they may need to be replaced, installed, and reconfigured – a process that could take weeks. If you have any questions or doubts, please contact the TAs for help.

Obstacle Course

The obstacle course is made up of a 4×6 grid of approximately 60-cm cubes, as shown below.



Each cube represents a known landmark, and is labeled on its bottom square face with a centered letter, **A**–**X**. Your robot is meant to move within the obstacle course without touching the walls. Note that if your robot significantly moves or damages any part of the obstacle course, your graded run will be paused to reset the course, and marks may be deducted.

Graded Runs

In Week 13, your team's robot will need to run on the obstacle course 3 times. Each run on the obstacle course will consist of the following steps:

1. Your team will be told:
   - A starting square S_0 and approximate bearing $B_0 \in \{\text{N, NE, E, SE, S, SW, W, NW}\}$.
   - An ordered sequence of four landmarks S_1, S_2, S_3, S_4 to visit (the landmarks are not necessarily distinct, i.e., you may be asked to visit the same landmark more than once).
   - Your team should program these inputs into your robot before each graded run, after which your team will not be allowed to interact with your robot until the next graded run.
2. The robot will be placed fully within square S_0 with bearing approximately B_0 ($\pm 45°$). Optionally, an obstacle may then be placed within some grid square, which will be visible on LiDAR and will block robot movement through that square. Your robot should not move or touch the obstacle.
3. Finally, the robot must autonomously navigate from S_0 to each landmark S_1, S_2, S_3, S_4 in order within a 3-minute time limit, stopping for at least 3 seconds at each landmark before proceeding.
4. The graded run will end after any of the following: (a) 3 minutes has elapsed, (b) the robot has traversed the course and stopped at S_4, or (c) the team interacts with the robot.

Scoring

Your graded runs collectively will be marked out of 50 points. Each of the 3 graded runs will be marked as follows. There are 4 paths the robot must traverse along each graded run sequence: the paths from S_{i-1} to S_i for $i \in \{1, 2, 3, 4\}$, where the start and end of each path is identified by stopping at the requested landmark for at least 3 seconds. Each path will be marked out of 4 points:

- $+1$: robot reaches above any part of square S_i
- $+1$: robot stops fully within S_i
- $+1$: robot stops in S_i while completely covering its center label
- $+1$: robot does not touch any wall or obstacle along the path to S_i from the previous landmark

If a robot fails to reach a landmark S_i, the team may elect to 'skip' it so as to be eligible for marks reaching a next landmark S_j for $j > i$, but will lose all marks for S_i, even if S_i is reached again after S_j. Skipping must be declared verbally by the team, prior to the reaching the next landmark S_j. So the maximum number of points a robot can score across 3 graded runs is $3 \times 4 \times 4 = 48$. One of the graded runs will be identical across all teams. The team that successfully completes that graded run the fastest (i.e., scores ≥ 3 points along every path) will receive an additional 2 points.

Mapping

In advance of the Week 13 evaluation, your team should learn to move and program your robot in the lab, and build an internal map of the obstacle course. It is important for your team to map the obstacle course early, so that your team has sufficient time to practice running the obstacle course in advance of Week 13. In pursuit of the goal, your robot may make use of any hardware on the platform (IMU, encoders, LiDAR, Camera, etc), and may use any software, but be advised that the TAs will only be able to help you debug libraries that we install for you.